

PREPARED FOR

GNSS Survey Report

39.15950, -122.15490

Section 14, T15N, R3W, Mount Diablo Meridian

✓ CORS Elev Delta: 118 ft above nearest CORS station (P270, 7.9 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
39.1595°N 122.1549°W WGS84	79 ft NAVD88 · MSL	CA 2 (0402) · N 2,184,128.52 E 6,517,745.77 (US ft, NAD83)	Zone X 06011C0517GG	13.1° E WMM 2026

APN 001-073-001-000 | ZONING 0-S | ACRES 1.909 ac

BEST GNSS WINDOW Wednesday Jul 15 · 2:00 PM – 6:00 PM | PDOP 0.94 · 35 sv | NEAREST CORS
P270 · 7.9 mi ✓

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 VIEW FIELD MAP ↗

 SITE SATELLITE ↗

 EMERGENCY MAP ↗

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER ASSESSOR
Colusa Colusa County
County Assessor · 546 Jay
Clerk- St, Colusa, CA
Recorder · 95932
· 546
Jay St,
Colusa,
CA 95932

NEAREST NEAREST BM
CORS KT0302 · 0.3 mi
P270 ·
7.9 mi

GENERATED SYSTEM
July 15, 2026 TopoScout ·
info@toposcout.com



This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.



INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

39.1595, -122.1549

WGS84 / NAD83

ELEVATION

79 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.1° E

Compass reads 13.1° too far W

FEMA FLOOD ZONE

X

0.2 PCT ANNUAL CHANCE FLOOD HAZARD · Panel 06011C0517GG

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC)

— UNAVAILABLE

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Wed, Jul 15	0.97	2:00 PM – 6:00 PM	35	Ideal
Thu, Jul 16	1.02	2:00 PM – 6:00 PM	33	Good
Fri, Jul 17	0.99	3:00 PM – 7:00 PM	34	Ideal
Sat, Jul 18	0.99	2:00 PM – 6:00 PM	35	Ideal
Sun, Jul 19	0.98	6:00 AM – 10:00 AM	33	Ideal
Mon, Jul 20	0.99	3:00 PM – 7:00 PM	34	Ideal
Tue, Jul 21	0.98	3:00 PM – 7:00 PM	33	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.97

►

RECOMMENDED WINDOW

Sun, Jul 19

5:00 AM – 9:00 AM

Avg wind 4 mph · Gusts 10 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Tue, Jul 21

10:00 AM – 2:00 PM

Avg wind 2 mph · Gusts 6 mph

Ideal conditions — 4-hr static occupation

🔥

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 15, 2026) · Conditions may change – verify before fieldwork

📶

GPS CONSTELLATION HEALTH

⚠️ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.0 across survey windows

PDOP remains ≤ 1.0 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 100°F

Low 70°F

Wind max 10 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:55 AM

Sunset 8:34 PM

New Moon 1%

GNSS: Avg PDOP **0.97** · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.94, 35 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.95	32 sv	IDEAL
7:00 AM	0.93	33 sv	IDEAL
8:00 AM	0.99	32 sv	IDEAL
9:00 AM	1.04	34 sv	GOOD
10:00 AM	0.93	33 sv	IDEAL
11:00 AM	1.04	26 sv	GOOD
12:00 PM	0.90	31 sv	IDEAL
1:00 PM	1.05	28 sv	GOOD
2:00 PM	0.94	34 sv	IDEAL
3:00 PM	0.94	36 sv	IDEAL
4:00 PM	0.99	32 sv	IDEAL
5:00 PM	0.88	36 sv	IDEAL
6:00 PM	1.03	31 sv	GOOD
7:00 PM	0.97	31 sv	IDEAL



GO – GOOD CONDITIONS

Clear sky | Wind: 9.6 mph (gusts 21.9 mph) | Precip: 0" | Temp: 70.2°–100°F



FIELD SAFETY WARNINGS

 HIGH HEAT 100°F — Monitor crew for heat stress



DRONE WIND ANALYSIS

 CAUTION Peak: 9.6/21.9 mph WNW

* Sunrise: 5:55 AM ◁ Sunset: 8:34 PM fly window clipped to daylight

 Best flight window: 6:00 AM – 7:00 PM (13 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	3/7 mph		NE
6 AM	2/8 mph		N
7 AM	4/11 mph		NW
8 AM	4/10 mph		NNW
9 AM	5/14 mph		N
10 AM	3/12 mph		N
11 AM	3/8 mph		NNE
12 PM	2/5 mph		NE
1 PM	2/5 mph		ENE
2 PM	4/8 mph		E
3 PM	6/11 mph		SE
4 PM	8/10 mph		SE
5 PM	7/10 mph		SE
6 PM	2/8 mph		SSW
7 PM	10/22 mph		WSW
8 PM	9/17 mph		WSW

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ► FREE TO FLY
Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107
→ TFR STATUS  NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 106°F

Low 61°F

Wind max 17 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:33 PM

New Moon 3%

GNSS: Avg PDOP **1.02** · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.98, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.93	32 sv	IDEAL
7:00 AM	1.01	32 sv	GOOD
8:00 AM	0.91	35 sv	IDEAL
9:00 AM	1.20	27 sv	GOOD
10:00 AM	1.07	28 sv	GOOD
11:00 AM	1.03	27 sv	GOOD
12:00 PM	0.95	30 sv	IDEAL
1:00 PM	1.05	29 sv	GOOD
2:00 PM	1.08	30 sv	GOOD
3:00 PM	0.99	31 sv	IDEAL
4:00 PM	1.00	34 sv	GOOD
5:00 PM	0.86	36 sv	IDEAL
6:00 PM	1.11	28 sv	GOOD
7:00 PM	1.13	27 sv	GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 16.5 mph (gusts 22.1 mph) | Precip: 0" | Temp: 61.4°–106°F



FIELD SAFETY WARNINGS

 EXTREME HEAT 106°F — Heat safety protocols required



DRONE WIND ANALYSIS

 CAUTION Peak: 16.5/22.1 mph S

* Sunrise: 5:56 AM

☾ Sunset: 8:33 PM

fly window clipped to daylight

✓ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	2/7 mph		ESE
6 AM	3/8 mph		ESE
7 AM	1/3 mph		N
8 AM	1/1 mph		S
9 AM	3/9 mph		SE
10 AM	5/8 mph		SSW
11 AM	6/6 mph		W
12 PM	4/4 mph		NW
1 PM	2/6 mph		SSW
2 PM	6/10 mph		SE
3 PM	10/12 mph		SE
4 PM	13/12 mph		SE
5 PM	13/11 mph		SSE
6 PM	15/12 mph		SSW
7 PM	17/18 mph		SW
8 PM	15/22 mph		SW

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 106°F

Low 61°F

Wind max 15 mph

Gusts 19 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:32 PM

Waxing Crescent 8%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	1.04		30 sv		GOOD
7:00 AM	1.12		27 sv		GOOD
8:00 AM	0.91		35 sv		IDEAL
9:00 AM	0.96		32 sv		IDEAL
10:00 AM	0.93		31 sv		IDEAL
11:00 AM	0.94		29 sv		IDEAL
12:00 PM	0.95		30 sv		IDEAL
1:00 PM	1.13		27 sv		GOOD
2:00 PM	1.03		31 sv		GOOD
3:00 PM	0.91		35 sv		IDEAL
4:00 PM	0.89		36 sv		IDEAL
5:00 PM	0.92		34 sv		IDEAL
6:00 PM	0.91		31 sv		IDEAL
7:00 PM	1.26		24 sv		GOOD

GO – GOOD CONDITIONS
Partly cloudy | Wind: 15.2 mph (gusts 19 mph) | Precip: 0" | Temp: 61°–105.5°F

FIELD SAFETY WARNINGS
 EXTREME HEAT 105.5°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 15.2/19 mph SSE

* Sunrise: 5:56 AM ⌂ Sunset: 8:32 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	2/5 mph		N
6 AM	4/6 mph		NNW
7 AM	4/9 mph		NNW
8 AM	5/8 mph		NNW
9 AM	4/6 mph		N
10 AM	3/4 mph		NNE
11 AM	1/3 mph		ESE
12 PM	2/4 mph		S
1 PM	4/7 mph		SSE
2 PM	7/11 mph		SE
3 PM	10/13 mph		SE
4 PM	11/13 mph		SE
5 PM	13/13 mph		S
6 PM	15/13 mph		S
7 PM	15/15 mph		S
8 PM	10/19 mph		S

BEST WINDOW
6:00 AM – 9:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ **FREE TO FLY**
Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#4

Saturday, Jul 18

WEEKEND

Clear sky

14 mph

GO

High 105°F Low 58°F Wind max 14 mph Gusts 20 mph Precip 0.00" Sunrise 5:57 AM Sunset 8:32 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.99 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.94, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.96	32 sv	IDEAL
7:00 AM	1.02	32 sv	GOOD
8:00 AM	1.03	32 sv	GOOD
9:00 AM	0.91	33 sv	IDEAL
10:00 AM	0.98	31 sv	IDEAL
11:00 AM	1.06	28 sv	GOOD
12:00 PM	0.96	30 sv	IDEAL
1:00 PM	1.01	31 sv	GOOD
2:00 PM	0.91	35 sv	IDEAL
3:00 PM	0.91	35 sv	IDEAL
4:00 PM	0.93	36 sv	IDEAL
5:00 PM	0.99	32 sv	IDEAL
6:00 PM	0.97	33 sv	IDEAL
7:00 PM	1.18	29 sv	GOOD



GO - GOOD CONDITIONS

Clear sky | Wind: 13.5 mph (gusts 19.5 mph) | Precip: 0" | Temp: 58.3°–105.1°F



FIELD SAFETY WARNINGS



EXTREME HEAT 105.1°F — Heat safety protocols required



DRONE WIND ANALYSIS



GO Peak: 13.5/19.5 mph SSE

* Sunrise: 5:57 AM Sunset: 8:32 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	5/4 mph		ESE
6 AM	2/4 mph		ESE
7 AM	1/7 mph		N
8 AM	2/7 mph		NNE
9 AM	3/7 mph		NNW
10 AM	2/6 mph		N
11 AM	1/3 mph		NNE
12 PM	2/3 mph		SSE
1 PM	4/5 mph		S
2 PM	6/8 mph		SSE
3 PM	8/10 mph		SE
4 PM	10/11 mph		SE
5 PM	11/11 mph		SSE
6 PM	13/11 mph		S
7 PM	14/12 mph		S
8 PM	11/20 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI Check B4UFLY ↗

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#5

Sunday, Jul 19

Overcast

7 mph

GO

WEEKEND

High 105°F

Low 66°F

Wind max 7 mph

Gusts 10 mph

Precip 0.00"

Sunrise 5:58 AM

Sunset 8:31 PM

☾ Waning Crescent 23%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		35 sv	IDEAL
7:00 AM	0.93		31 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	0.94		31 sv	IDEAL
10:00 AM	1.09		28 sv	GOOD
11:00 AM	1.14		25 sv	GOOD
12:00 PM	0.99		28 sv	IDEAL
1:00 PM	0.95		31 sv	IDEAL
2:00 PM	0.99		33 sv	IDEAL
3:00 PM	0.95		33 sv	IDEAL
4:00 PM	0.95		36 sv	IDEAL
5:00 PM	0.89		35 sv	IDEAL
6:00 PM	0.95		32 sv	IDEAL
7:00 PM	1.07		29 sv	GOOD

☀️

GO — GOOD CONDITIONS

Partly cloudy | Wind: 7.4 mph (gusts 10.3 mph) | Precip: 0" | Temp: 66.4°–105.2°F

⚠️

FIELD SAFETY WARNINGS

🌡️ EXTREME HEAT 105.2°F — Heat safety protocols required

🚁

DRONE WIND ANALYSIS

✅ GO Peak: 7.4/10.3 mph SSE

* Sunrise: 5:58 AM 🌄 Sunset: 8:31 PM fly window clipped to daylight

✅ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	4/5 mph		ENE
6 AM	3/8 mph		NNW
7 AM	4/10 mph		N
8 AM	4/8 mph		N
9 AM	6/10 mph		NNW
10 AM	6/8 mph		N
11 AM	6/7 mph		NNE
12 PM	4/5 mph		NNE
1 PM	1/3 mph		NNE
2 PM	2/2 mph		S
3 PM	3/3 mph		SSE
4 PM	5/5 mph		SE
5 PM	6/7 mph		SE
6 PM	7/9 mph		SSE
7 PM	7/10 mph		S
8 PM	7/10 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ✅ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 111°F

Low 69°F

Wind max 17 mph

Gusts 21 mph

Precip 0.00"

Sunrise 5:59 AM

Sunset 8:30 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 34 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		34 sv	IDEAL
7:00 AM	0.92		35 sv	IDEAL
8:00 AM	0.91		36 sv	IDEAL
9:00 AM	1.00		30 sv	GOOD
10:00 AM	0.98		28 sv	IDEAL
11:00 AM	1.22		24 sv	GOOD
12:00 PM	0.93		29 sv	IDEAL
1:00 PM	1.08		26 sv	GOOD
2:00 PM	1.02		32 sv	GOOD
3:00 PM	0.98		34 sv	IDEAL
4:00 PM	0.94		33 sv	IDEAL
5:00 PM	0.92		33 sv	IDEAL
6:00 PM	0.87		35 sv	IDEAL
7:00 PM	1.19		28 sv	GOOD

GO — GOOD CONDITIONS
Partly cloudy | Wind: 16.6 mph (gusts 21 mph) | Precip: 0" | Temp: 68.6°–110.7°F

FIELD SAFETY WARNINGS
 EXTREME HEAT 110.7°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 16.6/21 mph SSE

* Sunrise: 5:59 AM ☾ Sunset: 8:30 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 8:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	5/2 mph		ESE
6 AM	5/2 mph		ESE
7 AM	4/2 mph		ESE
8 AM	4/3 mph		ESE
9 AM	3/2 mph		ESE
10 AM	3/2 mph		ESE
11 AM	3/2 mph		SE
12 PM	4/4 mph		SE
1 PM	4/6 mph		SSE
2 PM	5/7 mph		SSE
3 PM	7/8 mph		SSE
4 PM	9/8 mph		SSE
5 PM	10/10 mph		SSE
6 PM	11/14 mph		S
7 PM	14/19 mph		SW
8 PM	17/21 mph		SW

BEST WINDOW
6:00 AM – 8:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ **FREE TO FLY**
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 113°F

Low 69°F

Wind max 15 mph

Gusts 19 mph

Precip 0.00"

Sunrise 5:59 AM

Sunset 8:30 PM

 Waxing Crescent 43%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.88		34 sv	IDEAL
7:00 AM	0.92		35 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	1.15		29 sv	GOOD
10:00 AM	1.01		31 sv	GOOD
11:00 AM	1.07		28 sv	GOOD
12:00 PM	0.99		29 sv	IDEAL
1:00 PM	1.06		28 sv	GOOD
2:00 PM	0.98		31 sv	IDEAL
3:00 PM	0.87		35 sv	IDEAL
4:00 PM	0.95		36 sv	IDEAL
5:00 PM	1.00		31 sv	GOOD
6:00 PM	0.96		30 sv	IDEAL
7:00 PM	1.02		29 sv	GOOD



GO – GOOD CONDITIONS

Partly cloudy | Wind: 14.7 mph (gusts 19 mph) | Precip: 0" | Temp: 69.3°–112.6°F



FIELD SAFETY WARNINGS



EXTREME HEAT 112.6°F — Heat safety protocols required



DRONE WIND ANALYSIS



GO Peak: 14.7/19 mph S

* Sunrise: 5:59 AM Sunset: 8:30 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	4/3 mph		ESE
6 AM	3/4 mph		E
7 AM	1/6 mph		NE
8 AM	1/7 mph		NNW
9 AM	2/6 mph		NNW
10 AM	2/5 mph		N
11 AM	2/5 mph		N
12 PM	1/9 mph		NNE
1 PM	2/6 mph		S
2 PM	4/7 mph		SSE
3 PM	6/7 mph		SE
4 PM	8/8 mph		SE
5 PM	9/10 mph		SE
6 PM	8/13 mph		S
7 PM	11/17 mph		SSW
8 PM	15/19 mph		SW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace – fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

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FIELD PLANNING REPORT — GNSS SURVEY OPERATIONS

Colusa County, CA | 39.1595, -122.1549

Generated: Wednesday, July 15, 2026 | Survey Start: July 15, 2026 | Analysis Window: 7 days

=====

TODAY (WEDNESDAY, JULY 15)

STATUS: GO

RATIONALE: Excellent PDOP geometry (avg 0.97, best 0.94 at 2–6 PM with 35 satellites). Clear skies, light winds (10 mph max), no precipitation. Moderate ionospheric activity (Kp 4.7 avg) acceptable for static work with standard occupation times. No satellite health impacts to constellation. Immediate mobilization recommended if crew available today; do not defer.

=====

MONDAY CALL (JULY 20) — CREW MOBILIZATION DECISION

STATUS: GO

RATIONALE: PDOP 0.99 average, best window 3–7 PM at PDOP 0.93 with 34 satellites. Overcast conditions do not degrade GNSS (no rain, 0.00" precipitation forecast). Wind 17 mph manageable. Moderate ionospheric activity continues. Crews should mobilize Monday with full confidence. This is a solid operational day.

=====

WEEK RANKING (1 = BEST, 7 = WORST)

1. FRIDAY, JULY 17 PDOP 0.99 avg, best 0.91 (3–7 PM, 34 sats)
2. SUNDAY, JULY 19 PDOP 0.98 avg, best 0.93 (6–10 AM, 33 sats)
3. TUESDAY, JULY 21 PDOP 0.98 avg, best 0.94 (3–7 PM, 33 sats)
4. WEDNESDAY, JULY 15 PDOP 0.97 avg, best 0.94 (2–6 PM, 35 sats) — TODAY
5. THURSDAY, JULY 16 PDOP 1.02 avg, best 0.98 (2–6 PM, 33 sats)
6. SATURDAY, JULY 18 PDOP 0.99 avg, best 0.94 (2–6 PM, 35 sats) [WEEKEND]
7. MONDAY, JULY 20 PDOP 0.99 avg, best 0.93 (3–7 PM, 34 sats)

NOTE: Top 5 are excellent work candidates. Weekend flagged for scheduling consideration. All days support survey-grade static and kinematic operations.

=====

BEST STATIC WINDOW (4-HOUR CONSECUTIVE SESSION)

FRIDAY, JULY 17, 3:00 PM – 7:00 PM UTC-7 (local time)

RATIONALE: Tightest PDOP window of the week (0.91), maximum satellite count (34), clear geometry across all constellations. Overcast conditions do not affect GNSS signal; no precipitation. Extend occupation time 10–15% to mitigate moderate ionospheric delay (Kp 4.7 avg) and ensure robust post-processing convergence.

=====

DRONE ASSESSMENT (MULTI-ROTOR / FIXED-WING)

Wednesday Jul 15: FLYABLE — clear sky, 10 mph wind, excellent visibility
Thursday Jul 16: CAUTION — 17 mph wind, clear sky; acceptable if within platform limits
Friday Jul 17: CAUTION — overcast, 15 mph wind; reduced visual line-of-sight risk
Saturday Jul 18: FLYABLE — clear sky, 14 mph wind, good visibility [WEEKEND]
Sunday Jul 19: FLYABLE — overcast, 7 mph wind (lightest day); reduced visual clarity
Monday Jul 20: CAUTION — overcast, 17 mph wind; marginal conditions
Tuesday Jul 21: CAUTION — overcast, 15 mph wind; monitor thermal stress (113°F high)

GUIDANCE: Prioritize Wednesday and Saturday for drone operations. Thursday and Tuesday marginal due to wind; Friday and Monday acceptable if platform rated for 15+ mph. Sunday ideal for low-wind work despite cloud cover.

=====

FIELD WARNINGS

HEAT HAZARD: Forecast highs 100–113°F; peak Tuesday. Enforce hydration protocols, frequent shade breaks, and equipment thermal management. RTK base station may require active cooling. Survey crews should start before 8 AM or after 5 PM on Tue–Wed.

WIND: Thursday (17 mph), Monday (17 mph), and Tuesday (15 mph) pose handling risk to tripods, antennas, and survey poles. Secure all equipment; consider guy-lines on mast.

SATELLITE MAINTENANCE: PRN 26 scheduled outage Fri 17 Jul 00:35–12:35 UTC will not affect afternoon static window (3–7 PM local = 22:35–02:35 UTC Thu–Fri). PRN 11 currently unhealthy (auto-excluded by receivers). PDOP scores already account for reduced constellation. No operational impact.

IONOSPHERIC ACTIVITY: Moderate Kp (4.7 avg) may induce 10–20 cm zenith delays in vertical component. Standard occupation times (2–4 hours for static baselines <10 km) sufficient; increase to 4+ hours if baseline >20 km or cm-level accuracy critical. TEC data unavailable; use default atmospheric models.

NO ACTIVE WILDFIRES: All-clear within reporting range; air quality not a constraint.

=====

OVERALL RECOMMENDATION

Begin survey operations TODAY (Wednesday) if mobilization feasible; do not defer work to later in week. Friday offers the week's tightest PDOP geometry (0.91) and should be prioritized for critical static sessions. Deploy crews Monday–Friday for maximum output; heat stress and wind escalate Tue–Wed, so front-load heavy fieldwork to earlier days and allocate lighter tasks (office, QC) to afternoon heat peaks. Moderate ionospheric activity and one unhealthy GPS satellite are manageable with standard protocols; no mission-critical constraints.

=====

END REPORT

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS
Colusa County Clerk-Recorder
546 Jay St, Colusa, CA 95932

ASSESSOR / TAX OFFICE
Colusa County Assessor
546 Jay St, Colusa, CA 95932

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P270	HOPKINSLGHCN2005	7.9 mi
 NGS ↗	SUTB	SUTTER BUTTES	18.2 mi
 NGS ↗	P336	HUBBARDRDGCN2007	29.4 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	KT0302	"Z"	24.050m / 78.9ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0435	"L"	26.120m / 85.7ft NAVD88	Order	0.4 mi
 Datasheet ↗	KT1857	"TIE"	32.000m / 105.0ft NAVD88	Order	0.5 mi
 Datasheet ↗	KT0433	"H"	27.810m / 91.2ft NAVD88	Order	0.5 mi
 Datasheet ↗	KT0304	"M 200 RESET"	23.520m / 77.2ft NAVD88	Order	0.7 mi
 Datasheet ↗	AA4332	"BLOCK RM"	26.010m / 85.3ft NAVD88	Order	0.8 mi
 Datasheet ↗	AA4365	"BLOCK RESET"	25.940m / 85.1ft NAVD88	Order	0.8 mi
 Datasheet ↗	KT0301	"BLOCK"	26.050m / 85.5ft NAVD88	Order	0.8 mi

FEMA FLOOD ZONE

X – 0.2 PCT ANNUAL CHANCE FLOOD HAZARD

Panel: 06011C0517GG
Effective: Mar 27, 2024
[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
KT0302	Z	39.158889	-122.149722	78.9	
KT0435	L	39.154017	-122.151025	85.7	
KT1857	TIE	39.166528	-122.150432	105.0	
KT0433	H	39.152500	-122.159444	91.2	
KT0304	M 200 RESET	39.168889	-122.156389	77.2	
AA4332	BLOCK RM	39.151242	-122.144611	85.3	
AA4365	BLOCK RESET	39.151167	-122.144528	85.1	
KT0301	BLOCK	39.151167	-122.144527	85.5	

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
SACRAMENTO R A COLUSA CA #11389500	9.1 mi	43.97 ft	8,640 cfs	● Normal
BEAR C AB HOLSTEN CHIMNEY CYN NR RUMSEY CA #11451715	17.2 mi	—	—	● Normal
CACHE C A RUMSEY CA #11451800	19.2 mi	2.90 ft	637 cfs	● Normal
SACRAMENTO R BL WILKINS SLOUGH NR GRIMES CA #11390500	20.5 mi	31.04 ft	7,120 cfs	● Normal

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	41.5 mi	4.92 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Oroville	44.4 mi	858.77 ft (-41 ft)	900 ft	2,826K af	80%
Black Butte Lake	45.9 mi	—	228 ft	73K af	53%
Folsom Lake	61.6 mi	446.81 ft (-19 ft)	466 ft	773K af	79%
Shasta Lake	108.6 mi	1,022.57 ft (-44 ft)	1,067 ft	3,350K af	74%
Lake Del Valle	110.2 mi	696.75 ft (-9 ft)	706 ft	36K af	67%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Whiskeytown Lake	102.3 mi	1,209 ft*	1,209 ft	238K af	99%
Trinity Lake	115.9 mi	2,370 ft*	2,370 ft	1,957K af	80%
New Melones Lake	120.9 mi	1,088 ft*	1,088 ft	1,659K af	69%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Black Butte-Pool	47 mi	456.6 ft	458.36 ft	—	100%
Englebright Lake-Pool	62 mi	525.1 ft	522.35 ft	—	101%
Camanche	101 mi	230.1 ft	221.86 ft	—	104%
Del Valle	110 mi	696.8 ft	695.01 ft	—	100%
Farmington Dam-Pool	120 mi	124.3 ft	117.09 ft	—	106%
Boca	143 mi	5,601.4 ft	5,579.88 ft	—	100%

SNOTEL STATIONS (NRCS) – within 50 miles

STATION	DIST	ELEV	SWE	DEPTH	TEMP	ACCESS
Noel Spring [Snow Course]	37.8 mi	4,950 ft	0"	—	—	● Clear

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
013N002W10K001M	12.5 mi	516 ft	—	—	S100CNRLVL
013N002W10L001M	12.5 mi	219 ft	112.00 ft BGS	1956-10-01	S100CNRLVL

013N002W11E002M	12.6 mi	256 ft	98.00 ft BGS	1954-10-01	S100CNRLVL
013N002W16A001M	13 mi	714 ft	—	—	S100CNRLVL
013N002W17N001M	13.2 mi	679 ft	—	—	S100CNRLVL
013N002W17Q001M	13.2 mi	749 ft	—	—	S100CNRLVL
013N001W07E001M	13.5 mi	168 ft	—	—	S100CNRLVL
013N002W15J001M	13.6 mi	362 ft	—	—	S100CNRLVL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Central Valley aquifer system (S100CNRLVL) in this Colusa County location consists of alluvial deposits laid down by ancient river systems and includes layers of sand, gravel, silt, and clay formed during the Quaternary Period. This mixed-layer system was deposited as the Sacramento River and its tributaries shifted course over thousands of years, creating a productive but heterogeneous groundwater resource. Water availability depends on the thickness and connectivity of permeable sand and gravel lenses; the well depths in your survey (168–749 ft) indicate drilling extends through multiple aquifer zones to access sustained yields.

FORMATION

Name	Central Valley alluvial aquifer (undifferentiated Quaternary)
Age / Rock	Quaternary alluvium and older flood-plain deposits; primarily sand, gravel, silt, and clay
Extent	8,000–12,000 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	0–800 ft (0–80 floors)
Typical Depth	100–200 ft (10–20 floors)
Recharge Source	Precipitation on regional uplands, lateral flow from Sierra Nevada foothills, and irrigation return flow
Transit Time	20–50 years to 150 ft depth

WATER QUALITY

pH	est. 6.8–7.6 (est. USGS/DWR)
Hardness	est. 180–380 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–700 mg/L (est. USGS/DWR)
Notes	No sample data available for this site. Central Valley alluvium typically exhibits moderate hardness and TDS; localized elevated nitrate possible in agricultural areas.

TREND

Trend	Water level declined 14.0 ft between October 1954 (98.0 ft BGS) and October 1956 (112.0 ft BGS), indicating sustained drawdown over 2-year period consistent with mid-20th century agricultural development
Source	Historical water level measurements (1954–1956)
Seasonal Swing	est. 8–20 ft seasonal swing (est. USGS/DWR)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The site near Lake County, California (39.1595, -122.1549) is located outside any critically overdrafted groundwater basin and overlies the S100CNRLVL aquifer system, which does not exhibit the excessive groundwater withdrawal patterns characteristic of subsidence-prone areas. The poorly drained soil classification indicates fine-grained materials with natural consolidation history, but the absence of intensive agricultural or industrial pumping in this region has prevented the anthropogenic aquifer depletion necessary to trigger compaction. For surveyors operating in this area, stable benchmarks and reliable elevation control are expected, though periodic verification of monument stability remains standard practice.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	93 mph
Risk Category	II	Ground Snow Load	6.9 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	6 in minimum
		Brown's Formula · C=1 (Clay, Group D) · AFI=9°F-days · SF=1.25	
		Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3B · Hot-Dry · Hot-Dry - warm/hot summers, cool winters, dry (Central CA, NV, AZ) County: Colusa

AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This hot-dry climate requires a well-insulated, air-tight envelope to minimize cooling loads, with particular attention to roof assembly since solar gain through the roof is your biggest challenge. Concentrate glazing on north and east facades while minimizing west and south-facing glass; any south or west windows must have deep overhangs or exterior shading to block summer sun. Your single most important passive strategy is to design deep roof overhangs (3–4 feet minimum) combined with strategic window placement to allow winter sun penetration on south walls while completely shading them in summer—this alone can reduce cooling energy by 20–30% in this climate.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This site is characterized by poorly drained silty clay soils with occasional flooding risk and very shallow frost depth. The combination of low bearing capacity potential, high water table, and occasional inundation may warrant a geotechnical investigation to assess foundation feasibility and anticipated settlement. Moderate wind exposure and modest snow loads are anticipated to be manageable with standard design practices.

FOUNDATION

- Type** Geotechnical investigation recommended before foundation selection — conventional spread footing may be challenged by soil drainage and flooding
- Basis** Willows silty clay (poorly drained, Hydro Group D) with occasional flooding anticipated — bearing capacity, settlement, and water management require site-specific evaluation

STRUCTURAL IMPLICATIONS

- Seismic** Seismic parameters null at this location — SDC classification cannot be determined from available data. Consult local building official to confirm seismic design category and any applicable detailing
- Wind Exposure** Exposure C (open terrain) anticipated per ASCE 7-22 — 93 mph 3-second gust wind speed suggested for planning
- Wind** Standard wind load design anticipated — no special exposure conditions (hurricane, special topography) noted. Verify exposure classification during design development
- Snow** Ground snow load of 6.9 psf anticipated — light to moderate snow load likely adequate with standard roof framing design

INVESTIGATIONS & INSPECTIONS

- Geotech Required** Yes — geotechnical investigation recommended to confirm bearing capacity, settlement potential, drainage characteristics, and fill/stabilization requirements given poor drainage and occasional flooding
- Liquefaction Screen** Screening recommended — Site Class D with poorly drained soils and inferred shallow groundwater may warrant liquefaction evaluation once seismic parameters are confirmed
- Special Inspections** Anticipated to depend on final SDC determination and foundation design — consult engineer and local building official
- Code Edition** ASCE 7-22 / IBC 2021 (planning reference only)
- Site Class Note** Site Class D assumed for planning based on soil description. Geotechnical investigation recommended to confirm classification, groundwater depth, and soil properties affecting foundation design

SOILS — USDA SOIL SURVEY

- MAP UNIT:** Willows silty clay, 0 to 1 percent slopes, occasionally flooded
- SERIES:** Willows
- DRAINAGE CLASS:** Poorly drained
- HYDRO GROUP:** D
- SOIL ORDER:** Vertisols
- SUBORDER:** Aquerts

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Willows series consists of poorly drained vertisols (clay-rich soils with high shrink-swell potential) formed in alluvial deposits on nearly level flood plains. The soil is characterized by a silty clay surface layer over clayey substratum with pronounced vertical cracks when dry and significant expansion when wet. Surveyors and developers should expect seasonal saturation, occasional flooding, severely restricted drainage, and substantial foundation movement risk due to clay mineralogy.

Alluvial parent material on level flood plains; poorly drained vertisols supporting sparse riparian vegetation.

DRAINAGE

- Class** Poorly drained
- Seasonal Saturation** Water table within 12–24 inches during winter and spring; saturated conditions common in subsoil year-round
- Flooding Risk** Occasional flooding on flood plains; risk increases in wet years
- Ponding Risk** Moderate — level terrain and poor drainage create ponding potential after heavy rainfall

ENGINEERING

- Bearing Capacity** 1,500–2,500 psf (shallow foundations); significantly reduced when saturated
- Rating** Poor
- Shrink-Swell** High — montmorillonitic clay content causes severe vertical movement (3–5+ inches annually possible)
- Frost Action** Low to moderate
- Cut Suitability** Fair — clay material is workable but prone to slaking and rutting when wet
- Fill Suitability** Poor — difficult to compact; moisture control essential; settlement risk high

SEPTIC

- Suitability** Unsuitable
- Notes** Seasonal high water table, poor drainage, and clay-rich substratum preclude septic system installation. Perc rates will be prohibitively slow or zero.

HAZARDS

Excavation

Moderate — clay becomes sticky and unstable when wet; equipment access severely limited during wet season

Expansive Clay

High — vertisol classification indicates substantial shrink-swell risk; structural cracking probable on foundations without special design

Erosion

Low on undisturbed slopes; however, disturbance to clay surface exposes material to erosion when saturated

Other

Seasonal access constraints due to ponding and saturation; flooding risk limits development suitability

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USBR RISE · [data.usbr.gov](#) · USACE CWMS · [water.usace.army.mil](#) · NRCS SNOTEL/AWDB · [wcc.sc.egov.usda.gov](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 15, 2026, 1:04 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

1	2	3
Endangered	Threatened	Total w/ Critical Habitat

1 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Vernal pool tadpole shrimp	<i>Lepidurus packardi</i>	Endangered
California tiger Salamander	<i>Ambystoma californiense</i>	Threatened
Yellow-billed Cuckoo	<i>Coccyzus americanus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Summary – 39.1595, -122.1549 ESA Section 7 consultation is triggered if federal permitting, funding, or agency involvement occurs, requiring a biological assessment to evaluate project effects on three species with critical habitat within 25 miles. The California tiger salamander (threatened) and vernal pool tadpole shrimp (endangered) present the most significant planning constraints due to their wetland habitat requirements and the likelihood of on-site or near-site occupation. The yellow-billed cuckoo (threatened) requires riparian habitat assessment if present. **Recommended next step:** Retain a qualified biologist to conduct a Phase 1 habitat assessment and determine whether listed species are present or likely to occur within the project footprint and vicinity; this will establish baseline conditions for permitting strategy and mitigation planning. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

55	84	2026-06-07
Unique Species	Total Observations	Most Recent

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Alkali Heath	<i>Frankenia salina</i>	2023-05-29
alkali heliotrope	<i>Heliotropium curassavicum</i>	2025-08-12
alkali mallow	<i>Malvella leprosa</i>	2023-09-02
Blow Wives	<i>Achyrachaena mollis</i>	2025-04-02
blue dicks	<i>Dipterostemon capitatus</i>	2018-03-19
blue elder	<i>Sambucus cerulea</i>	2022-04-17
blue oak	<i>Quercus douglasii</i>	2025-04-10
box elder	<i>Acer negundo</i>	2021-03-24
Buckbrush	<i>Ceanothus cuneatus</i>	2025-02-22
bull clover	<i>Trifolium fucatum</i>	2022-06-22
Bunchleaf Penstemon	<i>Penstemon heterophyllus</i>	2024-04-20
California buckeye	<i>Aesculus californica</i>	2022-05-07
California Buckwheat	<i>Eriogonum fasciculatum</i>	2021-08-11
California poppy	<i>Eschscholzia californica</i>	2023-04-17
California wild grape	<i>Vitis californica</i>	2019-10-11
California Wild Rose	<i>Rosa californica</i>	2019-09-28
Common Manzanita	<i>Arctostaphylos manzanita</i>	2026-01-24
creamcups	<i>Platystemon californicus</i>	2023-04-27
Erect burhead	<i>Echinodorus berteroi</i>	2023-09-02
False Daisy	<i>Eclipta prostrata</i>	2023-10-01
False Monkeyflower	<i>Mimetanthe pilosa</i>	2026-06-07
fivespot	<i>Nemophila maculata</i>	2024-03-18
fringed checkerbloom	<i>Sidalcea diploscypha</i>	2023-05-14

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Giant Blazingstar	<i>Mentzelia laevicaulis</i>	2024-07-28
harlequin calicoflower	<i>Downingia insignis</i>	2023-05-22
hooded arrowhead	<i>Sagittaria calycina</i>	2018-07-12
huckleberry oak	<i>Quercus vaccinifolia</i>	2020-06-10
jeweled onion	<i>Allium serra</i>	2024-04-14
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2023-09-02
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2023-09-02
pale smartweed	<i>Persicaria lapathifolia</i>	2023-09-02
purple owl's-clover	<i>Castilleja exserta</i>	2026-03-24
Rancher's fiddleneck	<i>Amsinckia menziesii intermedia</i>	2018-03-19
Red larkspur	<i>Delphinium nudicaule</i>	2020-06-10
rough cocklebur	<i>Xanthium strumarium</i>	2023-09-02
Sacred Datura	<i>Datura wrightii</i>	2025-04-07
scarlet monkeyflower	<i>Erythranthe cardinalis</i>	2026-06-01
scytheleaf onion	<i>Allium falcifolium</i>	2025-05-14
Sharpleaf groundcherry	<i>Physalis acutifolia</i>	2019-08-26
shy monkeyflower	<i>Erythranthe nasuta</i>	2025-04-11
slender stinging nettle	<i>Urtica gracilis</i>	2019-04-13
small fescue	<i>Festuca microstachys</i>	2023-05-14
sparse-flowered bog orchid	<i>Platanthera sparsiflora</i>	2020-06-10
swamp smartweed	<i>Persicaria hydropiperoides</i>	2026-03-02
tall flatsedge	<i>Cyperus eragrostis</i>	2025-07-03
Tiny Mousetail	<i>Myosurus minimus</i>	2025-04-02
turkey mullein	<i>Croton setiger</i>	2024-06-30
turkey tangle frogfruit	<i>Phyla nodiflora</i>	2025-07-04
twining snakelily	<i>Dichelostemma volubile</i>	2024-05-11
valley oak	<i>Quercus lobata</i>	2023-09-02
Virgate Phacelia	<i>Phacelia heterophylla virgata</i>	2020-06-01
water fern	<i>Azolla filiculoides</i>	2025-07-03
western maidenhair fern	<i>Adiantum aleuticum</i>	2020-06-10
woolly rosemallow	<i>Hibiscus lasiocarpus</i>	2025-07-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-05-02

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 39.1595, -122.1549 This site supports a **California oak woodland and chaparral transitional community**, dominated by drought-tolerant shrubs (buckbrush, manzanita, California buckwheat) and scattered oak species (blue oak, *Quercus douglasii*) typical of the inland Coast Ranges and Sierra foothills. The presence of alkali-adapted species (alkali heath, alkali heliotrope, alkali mallow) indicates seasonal wetland edges or alkaline soil conditions, suggesting the site occupies a moisture gradient where upland chaparral meets lower-elevation riparian or vernal pool influence. Grading activities should anticipate variable drainage: upland areas will have well-draining, potentially shallow soils supporting chaparral, while lower-lying zones may experience seasonal waterlogging or alkaline conditions requiring modified drainage design and fill material. These conditions—combined with the prevalence of native shrubs indicating fire-adapted habitat—necessitate careful erosion control and native vegetation retention during construction. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

24	6	1	51
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		24 obs	
Crotalus oreganus		24 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		6 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla sackenii		1 obs	

TOXIC PLANTS

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	19 obs
<i>Datura stramonium</i>	22 obs
<i>Datura ferox</i>	22 obs
<i>Datura wrightii</i>	22 obs
<i>Conium maculatum</i>	7 obs
<i>Urtica gracilis</i>	3 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Northern California Survey Site **Highest-Priority Hazards:** Northern Pacific rattlesnakes (*Crotalus oreganus*) are your primary concern with 24 documented observations in this region—they're present year-round but most active April through October. Poison oak (*Toxicodendron diversilobum*) is widespread (51 observations) and causes severe contact dermatitis in most people; Datura species and hemlock are also present and toxic if ingested. Black widows (*Latrodectus hesperus*) and other stinging insects pose secondary risks. **Practical Avoidance:** Stay on cleared survey paths, wear closed-toe boots and long pants, and watch for snakes before placing hands or feet in brush. Learn to identify poison oak (three-leaflet clusters) and avoid touching it; wash exposed skin immediately if contact occurs. Never eat or taste unfamiliar plants, and shake out equipment before packing. **Seasonal Note:** Spring through fall brings peak snake activity and plant toxin concentration; exercise heightened vigilance during warmer months. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

1	523ft	1		
Beam Paths	Closest Beam	Fixed P2P		
TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WNTP597	WNTP596	Fixed Point-to-Point	1.1 mi	523ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

23	8	6	9			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	342.8°	161.8°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J189	Jet Route	High	18,000 ft	—	335.5°	155.0°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	22,000 ft	—	342.6°	162.6°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	162.7°	343.3°
T259	T-Route	Low	—	—	340.5°	160.2°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	5,000 ft	—	288.9°	111.8°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°
V150	Victor Airway	Low	3,000 ft	—	35.4°	215.5°
V195	Victor Airway	Low	5,300 ft	—	352.9°	172.9°
V199	Victor Airway	Low	9,000 ft	—	21.1°	201.2°
V200	Victor Airway	Low	6,200 ft	—	72.5°	252.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	S	1	No	No	—
CFNR	Mainline	1	No	No	WEST VALLEY
CFNR	Other	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	85	68	7	7	3
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (68 species)

COMMON NAME	Scientific Name	OBS
Black-crowned Night Heron	Nycticorax nycticorax	9
Great Egret	Ardea alba	8
Western Cattle-Egret	Ardea ibis	8
Snowy Egret	Egretta thula	7
Western Kingbird	Tyrannus verticalis	6
Red-winged Blackbird	Agelaius phoeniceus	6
Double-crested Cormorant	Nannopterum auritum	6
Black-necked Stilt	Himantopus mexicanus	6
Killdeer	Charadrius vociferus	5
White-faced Ibis	Plegadis chihi	5
Wild Turkey	Meleagris gallopavo	4
American Bittern	Botaurus lentiginosus	4
House Sparrow	Passer domesticus	4
Greater White-fronted Goose	Anser albifrons	4
Ring-necked Pheasant	Phasianus colchicus	3

+ 53 additional species

MAMMALS (7 species)

COMMON NAME	Scientific Name	OBS
Columbian Black-tailed Deer	Odocoileus hemionus columbianus	4
Black-tailed Jackrabbit	Lepus californicus	3
American Badger	Taxidea taxus	2
Elk	Cervus canadensis	2
Coyote	Canis latrans	1
Douglas's Ground Squirrel	Otospermophilus douglasii	1
Tule Elk	Cervus canadensis nannodes	1

REPTILES (7 species)

COMMON NAME	Scientific Name	OBS
Pacific Gopher Snake	Pituophis catenifer catenifer	5
Northern Pacific Rattlesnake	Crotalus oreganus oreganus	5
Northwestern Pond Turtle	Actinemys marmorata	5
Western Fence Lizard	Sceloporus occidentalis	2
Western Yellow-bellied Racer	Coluber constrictor mormon	2
Striped Racer	Masticophis lateralis	2
California King Snake	Lampropeltis californiae	1

AMPHIBIANS (3 species)

COMMON NAME	Scientific Name	OBS
American Bullfrog	Lithobates catesbeianus	3
Western Spadefoot	Spea hammondi	2
Western Toad	Anaxyrus boreas	1

Ecological Character Assessment This site in the Lower Lake area of Lake County, California supports a wetland-associated wildlife community indicative of riparian and seasonal wetland habitats with adjacent upland grassland and scrub. The bird assemblage is dominated by wading birds and waterbirds (herons, egrets, ibis, cormorants, stilts, killdeer) reflecting shallow water and marsh conditions, while the presence of Western Spadefoot, Northwestern Pond Turtle, and American Bullfrog confirms seasonal or permanent aquatic features. The mammalian diversity (including both Columbian Black-tailed Deer and Tule Elk) and reptile community suggest a mosaic of grassland, oak woodland, and disturbed areas typical of foothill agricultural or rangeland contexts with moderate human activity. The species composition indicates the site functions as important waterfowl and waterbird habitat with potential sensitivity for special-status species including Tule Elk and wetland-dependent amphibians, suggesting that any development should incorporate wetland and riparian buffer protections and biological surveys prior to ground disturbance. For planning purposes only. Consult a qualified biologist for site-specific biological assessments.

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳ PASSIVE + ACTIVE SOLAR PLANNING — Lat 39.16°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 39.2° (latitude) → 10/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	96.9%	good
5/12	22.6°	97.5%	good
6/12	26.6°	98.1%	← excellent
7/12	30.3°	98.7%	← excellent
8/12	33.7°	99.2%	← excellent
9/12	36.9°	99.7%	← excellent
10/12	39.8°	99.9%	← annual optimal
12/12	45°	99.1%	← excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	49	30	Minimize south glass
South glass (SHGC 0.6)	73	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	26	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	234	301	Highest load surface
Roof — light (absorb 0.35)	86	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	61	79	Best roof option for heat
South wall — dark	91	69	
South wall — light	30	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3B Hot-Dry: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.2	2	2.4	OK

30ft	All year	3.1	2.9	3.8	Zone 2 check
40ft	All year	4.1	3.8	5.4	△ verify
50ft	All year	4.9	4.7	7.8	△ verify
60ft	All year	5.7	5.5	9.2	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🕒 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar and Energy Plan You're in a fantastic location for solar—you get about 6.6 peak sun hours daily on average, which is like having six and a half hours of perfect, full-strength sunlight hitting your panels every single day; in summer it jumps to nine hours, though winter dips to just under four, so a south-facing roof tilted at about 39 degrees will drink in that sunshine most efficiently year-round. Your roof color is surprisingly powerful: a dark roof absorbs summer heat like a cast-iron skillet left in the sun, pulling in about 300 BTUs per hour per square foot, while a light-colored roof cuts that nearly in half—so painting your roof light or choosing light shingles is almost as important as insulation for keeping cooling costs down. Watch out for west-facing windows, though, because that afternoon sun hammers them with about 100 BTUs per hour, making your home feel like an oven in summer; even good blinds help, but you might want to plan overhangs or landscaping there. Here's something clever: a big deciduous tree on the south side of your house will shade you for about 4 hours in summer (perfect!) while dropping its leaves in winter to let warm sun through (4+ hours of free heat)—though if you're in a wildfire zone, check with your local fire department about safe tree distances first, since fire safety rules vary. Thermal mass—basically heavy materials like concrete or brick inside your walls—acts like a thermal battery, soaking up heat during the day and slowly releasing it at night, smoothing out temperature swings so your HVAC doesn't work as hard. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing 39.15950°N, 122.15490°W | Colusa County, California

Structural Design Drivers

Wind and snow load data are not available for this site. Seismic demand cannot be assessed from the data provided. The absence of these critical load cases prevents determination of which parameter dominates the structural design envelope. The design team must obtain site-specific wind speed data (ASCE 7 Figure 26.5-1C for Colusa County) and confirm that seismic demand is either negligible or obtain USGS-compliant site-class spectral accelerations before selecting the primary load path. Until those inputs are captured, structural system selection cannot proceed on a defensible basis.

Geotechnical & Foundation Design

The site is underlain by Willows silty clay on slopes of 0 to 1 percent. This soil is classified Hydrologic Group D (poorly drained) with poor surface drainage characteristics. Group D soils typically exhibit high clay content, low permeability, and poor bearing capacity compared to Groups A and B; standard design values should not be assumed without site-specific geotechnical investigation. The site is subject to occasional flooding, which introduces saturation and lateral pressures on below-grade elements and requires stormwater management and potentially raised floor elevations or waterproofing.

Frost depth is 6 inches (calculated via Brown's Formula using an Accumulated Freezing Index of 9°F-days for IECC Zone 3B, Hot-Dry climate). Footings in this climate must extend to or below 6 inches below final grade or be designed to accommodate shallow frost heave; however, the poorly drained Group D soils are prone to heave when saturated and frozen. A site-specific geotechnical investigation must characterize in-situ moisture content, bearing capacity at design depth, and potential for frost heave under the water table or capillary rise. Shallow spread footings on Willows silty clay will require confirmation of bearing capacity and settlement analysis, and mat or pile foundations may be warranted if loads are heavy or future saturation is likely.

Hazard Assessment

No Quaternary faults are recorded within the search radius. No subsidence basin data exists for this location, indicating either historical stability or data gaps in the CalGIS subsidence record. No mineral deposits or active oil/gas wells are listed in MRDS or CalGEM databases within the search area. Wildfire risk data were not provided and should be obtained from CAL FIRE for Colusa County. Liquefaction potential cannot be assessed without groundwater depth and cyclic soil testing; given the poorly drained nature of the site and occasional flooding, water table elevation must be determined.

Site Context & Climate Envelope

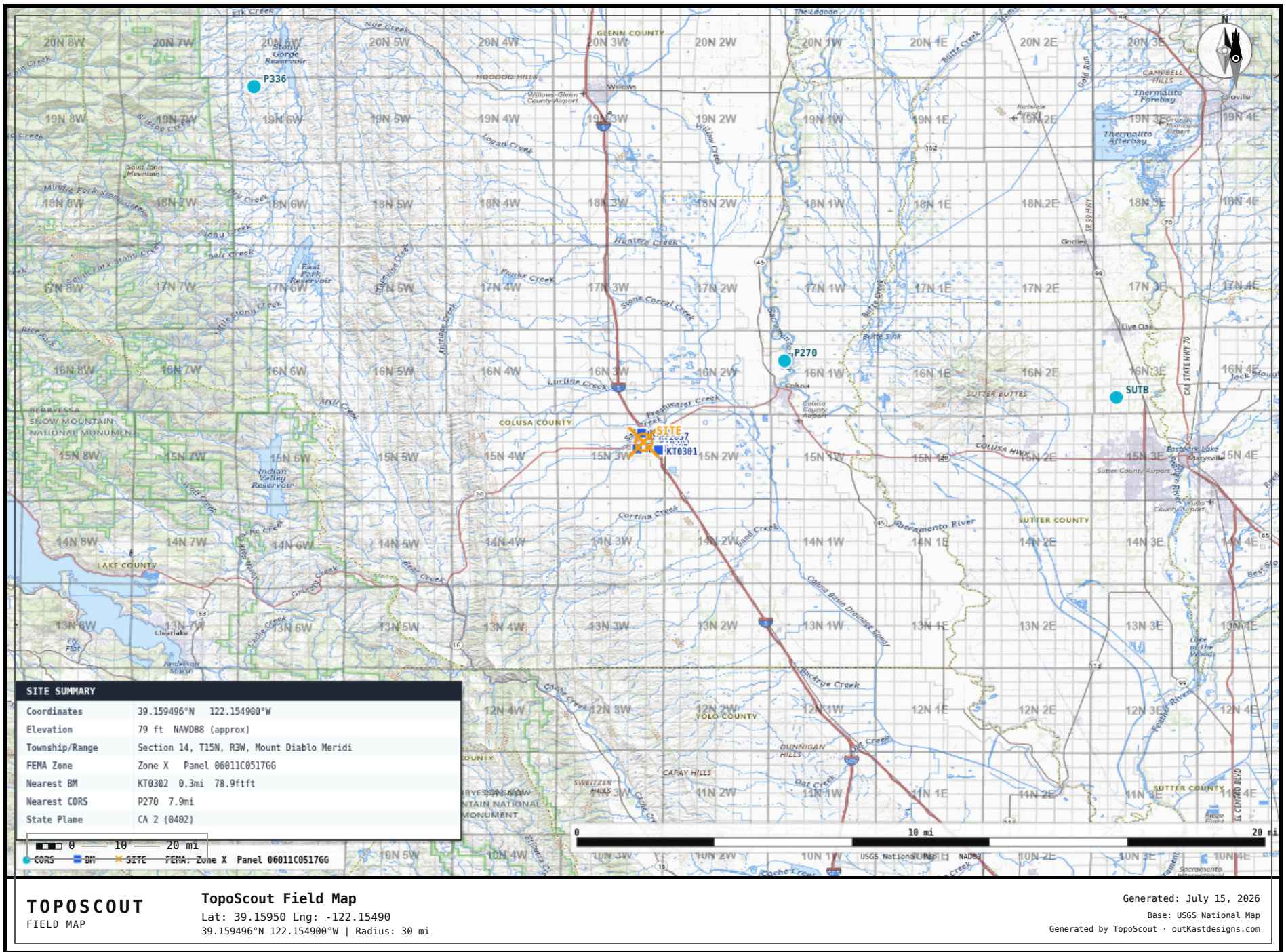
The site is located in IECC Climate Zone 3B (Hot-Dry) with an AFI of 9°F-days, establishing minimal freezing demand for mechanical systems and envelope design. The hot-dry designation confirms low annual precipitation in Colusa County, consistent with the Sacramento Valley agricultural region, though occasional flooding on this specific parcel indicates local drainage constraints or seasonal overflow from adjacent watercourses. No proximity to active mines or subsidence zones simplifies mineral-related hazards.

Bottom Line: Obtain site-specific seismic demand (USGS Site Class and MCE spectral accelerations), wind speed data, and a Phase I geotechnical investigation including soil boring, water table depth, and bearing capacity testing before finalizing the structural system and foundation design.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.



TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polygons (all geometry)



SITE AERIAL – 600 ft × 600 ft

39.15950°N, 122.15490°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

39.15950°N, 122.15490°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

